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Regulatory Policy and Practice on AI's Frontier

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Adaptive, expert-led regulation can unlock the promise of artificial intelligence.

Technological breakthroughs, historically, have played a distinctive role in accelerating economic growth, expanding opportunity, and enhancing standards of living. Technology enables us to get more out of the knowledge we have and prior scientific discoveries, in addition to generating new insights that enable new inventions.

Technology is associated with new jobs, higher incomes, greater wealth, better health, educational improvements, time-saving devices, and many other concrete gains that improve people's day-to-day lives. The benefits of technology, however, are not evenly **distributed**, even when an economy is more productive and growing overall. When technology is disruptive, costs and dislocations are shouldered by some more than others, and periods of transition can be difficult.

Theory and experience **teach** that innovative technology does not automatically improve people's station and situation merely by virtue of its development. The way technology is deployed and the degree to which gains are shared—in other words, turning technology's promise into reality without overlooking valid concerns—depends, in meaningful part, on the policy, regulatory, and ethical decisions we make as a society.

Today, these decisions are front and center for artificial intelligence (AI).

AI's capabilities are remarkable, with profound implications **spanning** health care, agriculture, financial services, manufacturing, education, energy, and beyond. The latest research is demonstrably **pushing** AI's frontier, advancing AI-based reasoning and AI's performance of complex multistep tasks, and bringing us closer to **artificial general intelligence** (high-level intelligence and reasoning that allows AI systems to autonomously perform highly complex tasks at or beyond human capacity in many diverse instances and settings). Advanced AI systems, such as AI agents (AI systems that autonomously complete tasks toward identified objectives), are leading to fundamentally new opportunities and ways of doing things, which can unsettle the status quo, possibly leading to major transformations.

In our view, AI should be embraced while preparing for the change it brings. This includes recognizing that the pace and magnitude of AI breakthroughs are faster and more impactful than anticipated. A terrific indication of AI's promise is the 2024 Nobel Prize in chemistry, winners of which **used** AI to “crack the code” of protein structures, “life's ingenious chemical tools.” At the same time, as AI becomes widely used, guardrails, governance, and oversight should manage risks, safeguard values, and look out for those disadvantaged by disruption.

Government can help fuel the beneficial development and deployment of AI in the United States by shaping a regulatory environment conducive to AI that fosters the

adoption of goods, services, practices, processes, and tools leveraging AI, in addition to encouraging AI research.

It starts with a pro-innovation policy agenda. Once the goal of promoting AI is set, the game plan to achieve it must be architected and implemented. Operationalizing policy into concrete progress can be difficult and more challenging when new technology raises novel questions infused with subtleties.

Regulatory agencies that determine specific regulatory requirements and enforce compliance play a significant part in adapting and administering regulatory regimes that encourage rather than stifle technology. Pragmatic regulation compatible with AI is instrumental so that regulation is workable as applied to AI-led innovation, further unlocking AI's potential. Regulators should be willing to allow businesses flexibility to deploy AI-centered uses that challenge traditional approaches and conventions. That said, regulators' critical mission of detecting and preventing harmful behavior should not be cast aside. Properly calibrated governance, guardrails, and oversight that prudently handle misuse and misconduct can support technological advancement and adoption over time.

Regulators can achieve core regulatory objectives, including, among other things, consumer protection, investor protection, and health and safety, without being anchored to specific regulatory requirements if the requirements—fashioned when agentic and other advanced AI was not contemplated—are inapt in the context of current and emerging AI.

We are not implying that vital governmental interests that are foundational to many regulatory regimes should be jettisoned. Rather, it is about how those interests are best achieved as technology changes, perhaps dramatically. It is about regulating in a way that allows AI to reach its promise and ensuring that essential safeguards are in place to protect persons from wrongdoing, abuses, and harms that could frustrate AI's real-world potential by undercutting trust in—and acceptance of—AI. It is about fostering a regulatory environment that allows for constructive AI-human collaboration—including using AI agents to help monitor other AI agents while humans remain actively involved addressing nuances, responding to an AI agent's unanticipated performance, engaging matters of greatest agentic AI uncertainty, and resolving tough calls that people can uniquely evaluate given all that human judgment embodies.

This takes modernizing regulation—in its design, its detail, its application, and its clarity—to work, very practically, in the context of AI by [accommodating](#) AI's capabilities.

Accomplishing this type of regulatory modernity is not easy. It benefits from combining technological expertise with regulatory expertise. When integrated, these dual perspectives assist regulatory agencies in determining how best to update regulatory frameworks and specific regulatory requirements to accommodate expected and unexpected uses of advanced AI. Even when underpinning regulatory goals do not change, certain decades-old—or newer—regulations may not fit with today's technology, let alone future technological breakthroughs. In addition, regulatory updates may be justified in light of regulators' own [use](#) of AI to improve regulatory processes and practices, such as [using](#) AI agents to streamline permitting, licensing, registration, and other types of approvals.

Regulatory agencies are filled with people who bring to bear valuable experience, knowledge, and skill concerning agency-specific regulatory domains, such as financial services, antitrust, food, pharmaceuticals, agriculture, land use, energy, the environment, and consumer products. That should not change.

But the commissions, boards, departments, and other agencies that regulate so much of the economy and day-to-day life—the administrative state—should have more technological expertise in-house relevant to AI. AI's capabilities are materially increasing at a rapid clip, so staying on top of what AI can do and how it does it—including understanding leading AI system architecture and imagining how AI might be deployed as it advances toward its frontier—is difficult. Without question, there are individuals across government with impressive technological chops, and regulators have made commendable strides keeping apprised of technological innovation. Indeed, certain parts of government are inherently technology-focused. Many regulatory agencies are not, however; but even at those agencies, in-depth understanding of AI is increasingly important.

Regulatory agencies should [bring](#) on board more individuals with technology backgrounds from the private sector, academia, research institutions, think tanks, and elsewhere—including computer scientists, physicists, software engineers, AI researchers, cryptographers, and the like.



For example, we envision a regulatory agency's lawyers **working** closely with its AI engineers to ensure that regulatory requirements contemplate and factor in AI. Lawyers with specific regulatory knowledge can **prompt** large language models to measure a model's interpretation of legal and regulatory obligations. Doing this systematically and with a large enough sample size requires close collaboration with AI engineers to automate the analysis and benchmark a model's results. AI engineers could partner with an agency's regulatory experts in discerning the technological capabilities of frontier AI systems to comport with identified regulatory objectives in order to craft regulatory requirements that account for and accommodate the use of AI in consequential contexts. AI could accelerate various regulatory functions that typically have taken considerable time for regulators to perform because they have demanded significant human involvement. To **illustrate**, regulators could use AI agents to assist the review of permitting, licensing, and registration applications that individuals and businesses must obtain before engaging in certain activities, closing certain transactions, or marketing and selling certain products. Regulatory agencies could augment humans by using AI systems to conduct an initial assessment of applications and other requests against regulatory requirements.

The more regulatory agencies have the knowledge and experience of technologists in-house, the more understanding regulatory agencies will gain of cutting-edge AI. When that enriched technological insight is combined with the breadth of subject-matter expertise agencies already possess, regulatory agencies will be well-positioned to modernize regulation that fosters innovation while preserving fundamental safeguards. Sophisticated technological know-how can help guide regulators' decisions concerning how best to revise specific regulatory features so that they are workable with AI and conducive to technological progress. The technical elements of regulation should be informed by the technical elements of AI to ensure practicable alignment between regulation and AI, allowing AI innovation to flourish without incurring undue risks.

With more in-house technological expertise, we think regulatory agencies will grow increasingly comfortable making the regulatory changes needed to accommodate, if not accelerate, the development and adoption of advanced AI.

There is more to technological progress that **propels** economic growth than technological capability in and of itself. An administrative state that is responsive to the capabilities of AI—including those on AI's expanding frontier—could make a big

difference converting AI's promise into reality, continuing the history of technological breakthroughs that have improved people's lives for centuries.



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